

Tomas Pereira - Data Scientist

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PROFILE

Computer Science graduate with hands-on experience in developing and deploying machine learning models and neural networks, managing large datasets, and building scalable systems. Strong background in deep learning, natural language processing, and reinforcement learning, with applied work in healthcare AI, multimodal models, and generative systems. Adept at bridging research and engineering, taking prototypes into production-ready solutions. Fully bilingual in English and French.

WORK EXPERIENCE

Master's/NSERC Researcher - Atlas Analytics Lab

May. 2023– Dec 2023

- Built multimodal vision-language models for automatic medical report generation using the CLIP framework.
- Designed a framework for histopathology object extraction using Meta's SAM model, improving preprocessing efficiency on large-scale datasets.
- Collaborated with medical experts at CHUM Hospital to align technical solutions with clinical requirements.
- Recipient of the NSERC-USRA Research Award (2023).

Teaching Assistant (Machine Learning) - Concordia University

Sept. 2023 - May 2025

- Instructed students in machine learning theory and implementation using PyTorch, NumPy, and Scikit-Learn.
- Mentored students in developing deep learning projects from conception to deployment.
- Delivered weekly labs and graded theoretical and applied assignments across multiple semesters.

Systems Specialist - Bureau Veritas

July. 2022 – May. 2023

- Automated data wrangling pipelines, reducing QA analyst record vetting time by 50%.
- Refactored and optimized VBA macros and datasheets to improve lab operations reliability.
- Key contributor to the rollout of a digital inventory system for lab supplies and biological samples.

EDUCATION

Master of Computer Science (MAPCompSc) - Concordia University

Sept. 2023 - May 2025

- Specialization in AI, LLMs, Reactive Programming, and Client-Server Applications
- Recipient of the Concordia Merit Scholarship Award (2024, 2025)
- Studied Representation Learning and Reinforcement Learning at MILA - Quebec AI Institute

Bachelor of Computer Science (BCompSc) - Concordia University

Sept 2019 - May 2023

- Recipient of the Government of Quebec Award under the Merit Scholarship Program (2021)

PROJECTS

MultiSpeech - Transformer-based Multi-Speaker Text-to-Speech Model

- Implemented and Trained a 7M-parameter TTS Transformer on the VCTK Dataset
- Generated Mel Spectrograms converted to audio with the HiFi-GAN vocoder
- Integration of complex attention mechanisms to enable the model to learn high quality multi-modal alignment.

Generative AI for Code Comment Updates

- Developed models to automatically update code comments after code changes.
- Fine-tuned CodeT5 and Google Gemma (LLM), improving BLEU and METEOR benchmarks.

Reinforcement Learning for WSI Analysis

- Designed an RL agent for efficient segmentation of whole-slide histopathology images.
- Reduced preprocessing costs while maintaining state-of-the-art classification accuracy.

PLAY Framework Reactive System

- Built a reactive application integrating the YouTube API with asynchronous message passing.
- Designed ETL pipeline for processing and delivering YouTubeV3 API results to clients.
- Implemented sentiment analysis on video descriptions.

SKILLS

- **Programming Languages:** Python, Java, SQL, VBA, C++
- **ML:** Pytorch, Numpy, Pandas, SKLearn, HuggingFace, OpenAI API, SpeechBrain, OpenCV
- **Systems and Tools:** SLURM, Docker, Linux, GIT
- **Specialties:** Deep Learning, Computer Vision, NLP, LLMs, RL, Computer Vision, Medical Imaging, Data Pipelines
- **Soft Skills:** Problem Solving, Critical Thinking, Collaboration, Mentorship, Communication, Innovation,